

Daniel Kwon-Reyes, Ph.D.

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RESEARCH INTERESTS

Statistical inference for high-dimensional biological and behavioral data; Bayesian hierarchical modeling; representation learning and evaluation of language models on scientific text. Increasingly interested in applying these methods to product measurement problems, where the estimand is defined by a design decision rather than by nature.

EDUCATION

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| 2016 – 2021 | Ph.D., Computational Neuroscience — University of California, Berkeley
<i>Dissertation:</i> "Hierarchical Bayesian Models of Population Coding Under Uncertainty."
Advisor: Prof. E. Mattingly. Redwood Center for Theoretical Neuroscience. |
| 2012 – 2016 | B.S. Physics and B.A. Statistics — University of Chicago
Phi Beta Kappa. Undergraduate thesis in statistical signal processing. |

RESEARCH & INDUSTRY APPOINTMENTS

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| 2024 – present | Senior Research Scientist, Machine Learning — insitro, South San Francisco, CA <ul style="list-style-type: none">• Lead statistician on a phenotype-prediction program: designed the hierarchical model that pools evidence across cell lines and batches, replacing per-batch point estimates that were driving irreproducible hit calls.• Built an internal evaluation harness for LLM-assisted literature triage — defined the annotation protocol, inter-rater agreement procedure (Krippendorff's α), and the held-out benchmark of 1,800 abstracts now used to gate model updates.• Designed a randomized internal trial comparing assisted and unassisted curation workflows across 22 scientists; the assisted arm reduced median triage time 41% with no measurable loss in precision.• Present quarterly to the research leadership team; author of the internal guidance on multiple-comparison control adopted across three programs. |
| 2021 – 2024 | Research Scientist — Recursion Pharmaceuticals, Salt Lake City, UT / remote <ul style="list-style-type: none">• Developed representation-learning methods for high-content cellular imaging (contrastive pretraining, batch-effect correction); models deployed in the primary screening pipeline.• Built the experimental-design toolkit used by wet-lab teams for plate randomization and power calculation; adopted by ~60 scientists.• Co-led an internal working group on reproducibility that introduced pre-registration for confirmatory screens. |
| 2019 | Research Intern, Machine Perception — Google Research, Mountain View, CA
Uncertainty calibration for sequence models; work contributed to publication [4]. |

SELECTED PUBLICATIONS

1. Kwon-Reyes, D., Aguilar, S., & Mattingly, E. (2024). Pooling across batches without pooling across biology: a hierarchical model for high-content screens. *Nature Methods*, 21(4), 612–621.
2. Iyer, R., Kwon-Reyes, D., & Chen, W. (2023). Evaluating language models on scientific abstract triage: benchmarks and failure modes. *Proceedings of NeurIPS Datasets & Benchmarks Track*.
3. Kwon-Reyes, D. & Mattingly, E. (2022). Uncertainty-aware population decoding in sensory cortex. *Journal of Neuroscience*, 42(19), 3810–3824.
4. Novak, P., Kwon-Reyes, D., et al. (2021). Calibration under distribution shift for sequence classifiers. *ICML*.
5. Kwon-Reyes, D., Ostrowski, J., & Mattingly, E. (2020). A variational approach to hierarchical spike-train models. *Journal of Machine Learning Research*, 21(88).
6. Aguilar, S. & Kwon-Reyes, D. (2019). Batch effects are a design problem, not a modeling problem. *Cell Systems* (Commentary), 9(3).

Full list of 19 publications available at dkwonreyes.github.io/publications.

INVITED TALKS

2025	"What experimental design owes to the people reading the result." Bay Area Machine Learning Symposium.
2024	"Hierarchical models for noisy screens." Broad Institute Models, Inference & Algorithms seminar.
2023	"Benchmarks that do not survive contact with users." NeurIPS D&B workshop, spotlight.

TECHNICAL SKILLS

Programming	Python (expert; 10 yrs), R, C++, Julia; SQL (working — Postgres, BigQuery; comfortable with joins, CTEs, window functions, less experience with production warehouse modeling)
Libraries	PyTorch, JAX, NumPyro, Stan, PyMC, scikit-learn, pandas, statsmodels, Hugging Face Transformers
Statistics	Bayesian hierarchical modeling, variational inference, MCMC, causal graphs, randomization and blocking, multiple-comparison control, power analysis, survival analysis
Infrastructure	Slurm, Docker, AWS Batch, Weights & Biases, Git; some Airflow
Languages	English (native), Korean (fluent), Spanish (fluent)

TEACHING & SERVICE

2022 – present	Reviewer: NeurIPS, ICML, <i>Nature Methods</i> , <i>Journal of Neuroscience</i> .
2018 – 2020	Graduate Student Instructor, UC Berkeley — Statistical Inference (Stat 135), Machine Learning (CS 189). Outstanding GSI Award, 2019.
2021 – present	Volunteer instructor, Bay Area Data Science for Public Health workshop series.

AWARDS

2017 – 2020	NSF Graduate Research Fellowship.
2021	Berkeley Neuroscience Dissertation Prize.